

Warehouse Workforce Forecasting & Labor Optimization Decision System

Using retail demand proxy data to improve staffing efficiency

Business Problem	Demand Volatility Creates Staffing Risk	Demand Patterns	Demand Thresholds.
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Reducing Reactive Staffing with Forecast-Driven VET/VTO Decisions

**Audience:**
Ops Managers / Workforce Planning

**Problem:**
Reactive staffing creates cost inefficiency

**Solution:**
Forecast demand + recommend staffing action



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Demand Volatility Creates Staffing Risk

Weekly retail demand is used as a proxy for warehouse workload. Demand is not stable across the year: some periods show clear spikes, while others fall below normal operating levels.

These patterns create staffing risk. Under-forecasting demand may lead to last-minute VET, while over-forecasting demand may lead to VTO inefficiency.

This project uses forecasting to convert demand patterns into earlier VET, VTO, or Normal planning signals.

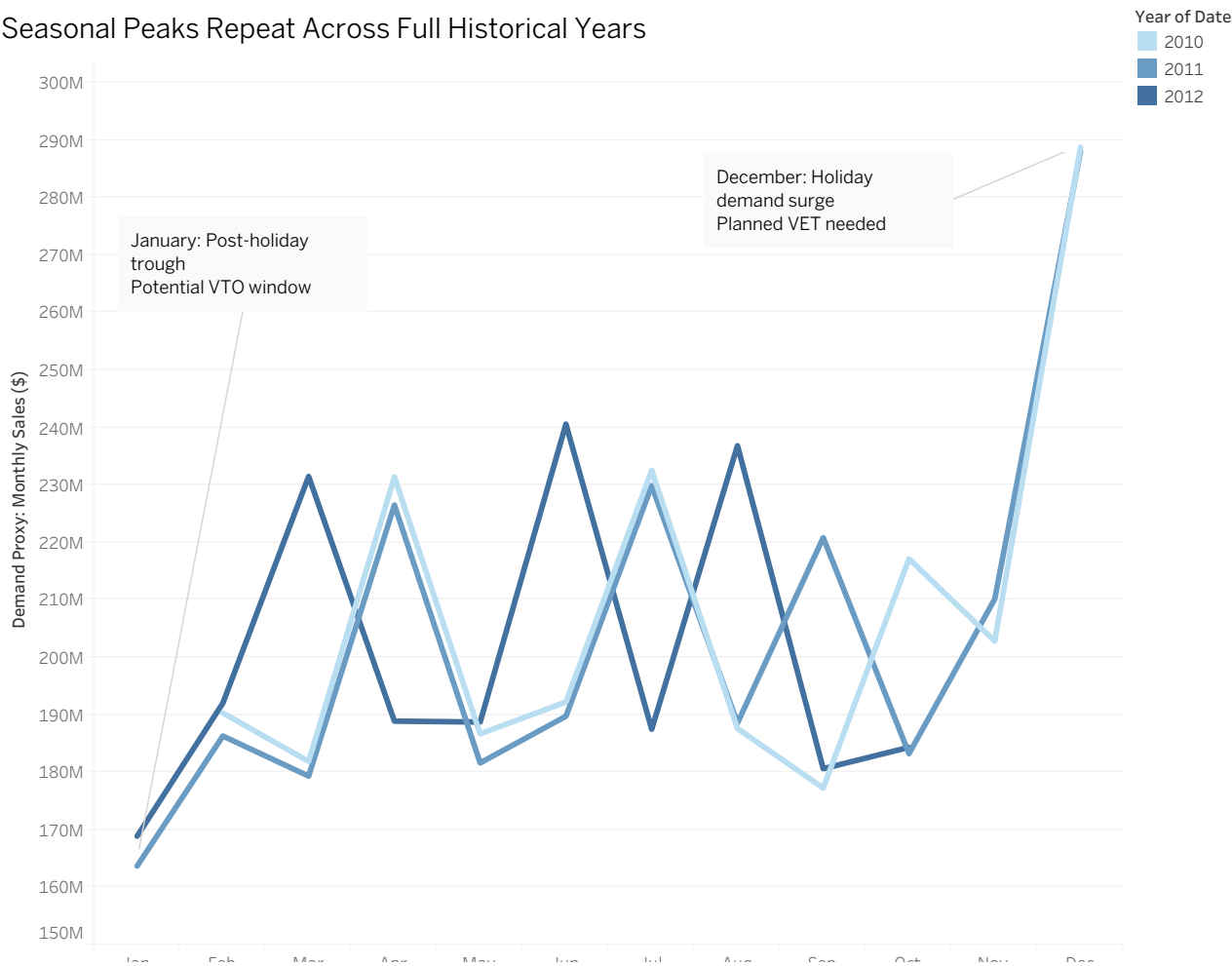
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Business Problem	Demand Volatility Creates Staffing Risk	Demand Patterns	Demand Thresholds.	Holiday Drivers
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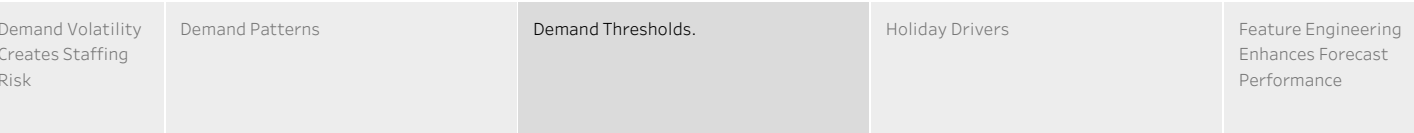
Repeating year-end peaks indicate demand can be anticipated, enabling proactive VET staffing decisions.

Seasonal Peaks Repeat Across Full Historical Years



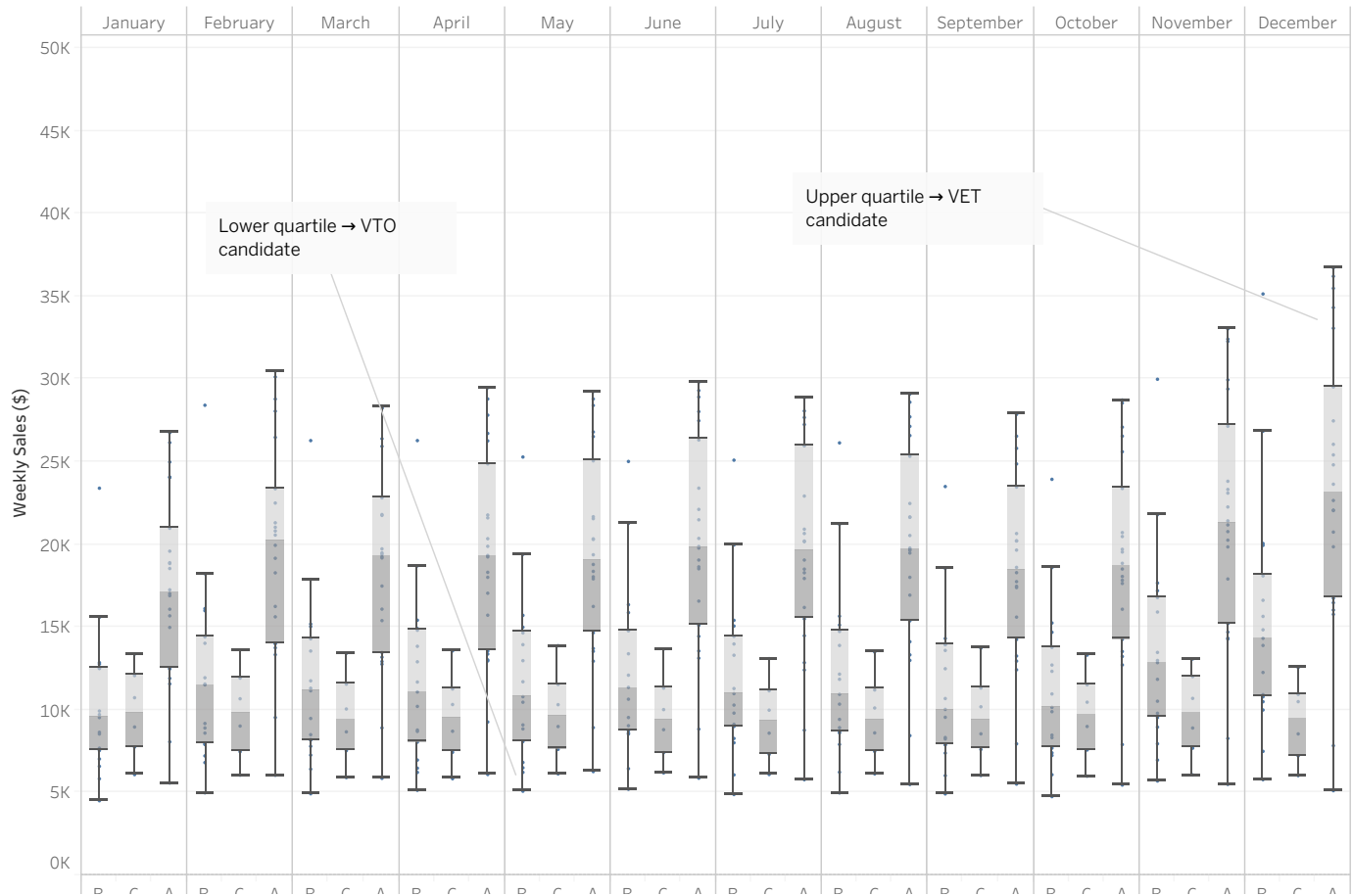
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Demand variability supports percentile thresholds for proactive staffing decisions.

Monthly Demand Distribution Reveals VET and VTO Windows



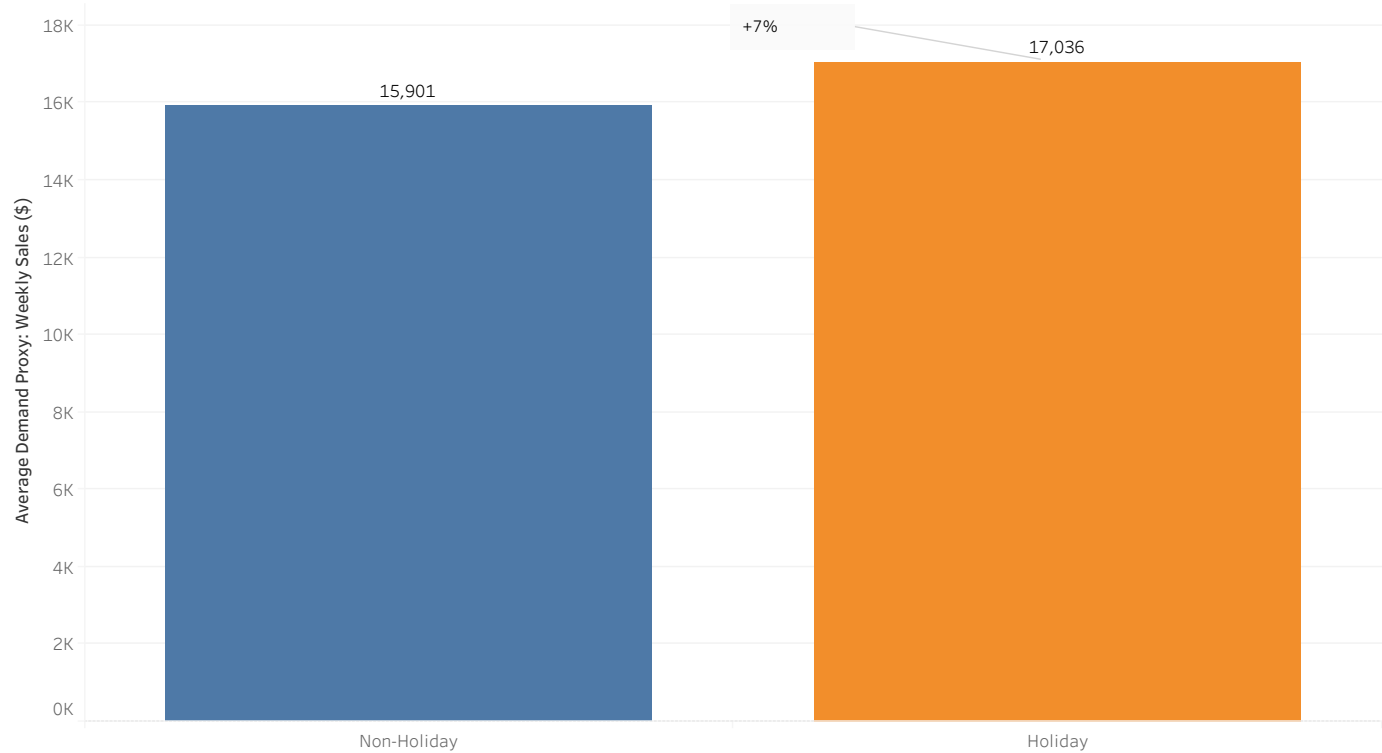
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Demand Patterns	Demand Thresholds.	Holiday Drivers	Feature Engineering Enhances Forecast Performance	XGBoost delivers the most accurate forecasts
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Holiday periods consistently increase demand, enabling advance staffing plans.

Holiday Periods Increase Demand by 7%



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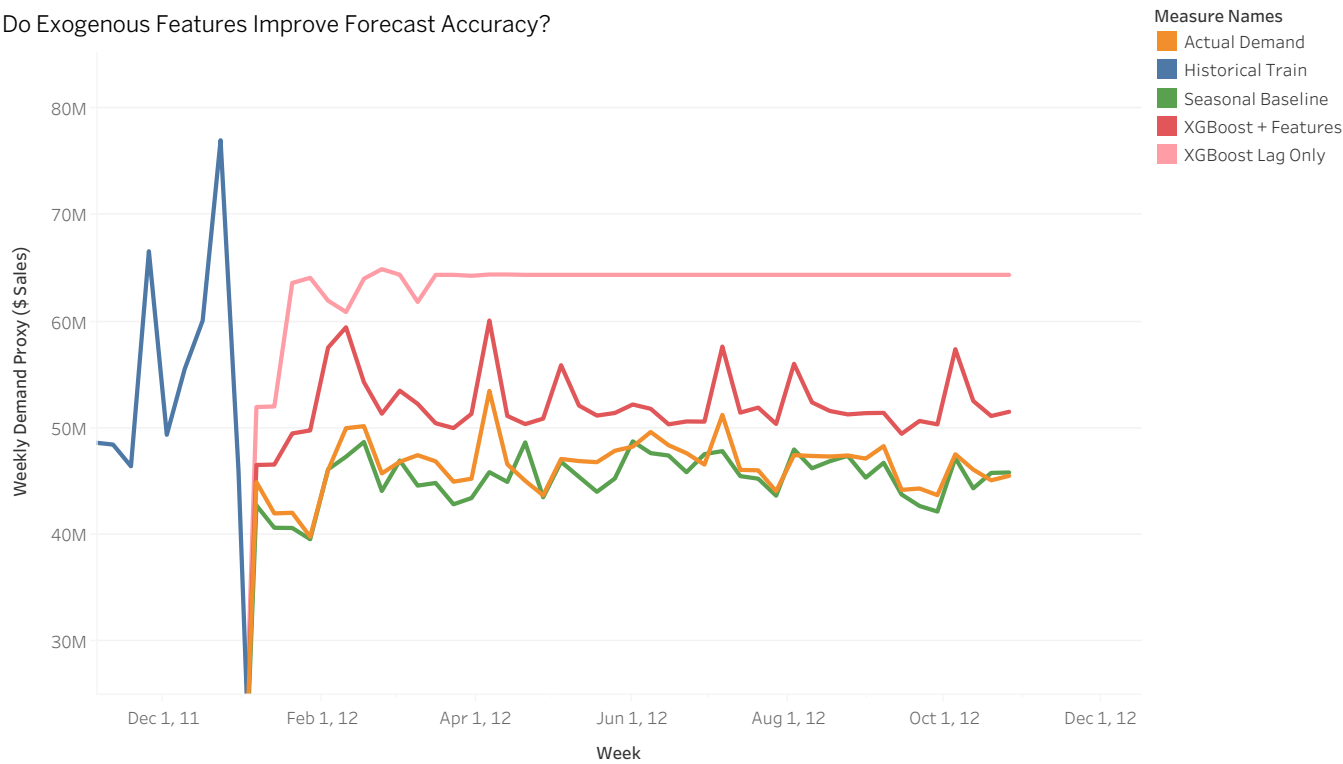
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Demand Thresholds.	Holiday Drivers	Feature Engineering Enhances Forecast Performance	XGBoost delivers the most accurate forecasts	XGBoost delivers reliable and accurate forecasts over time
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Feature Engineering

- Lag features capture recurring demand cycles
- Rolling metrics detect short-term momentum
- Holidays and macro variables add context
- Proxy features approximate operational stress signals

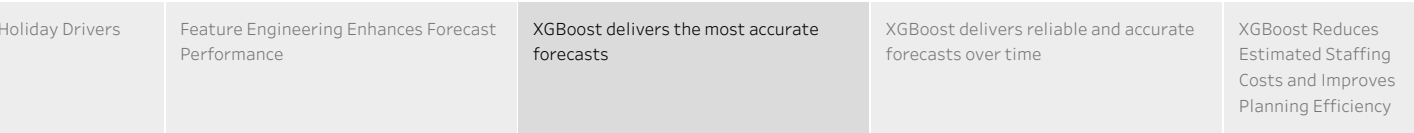
Do Exogenous Features Improve Forecast Accuracy?



Feature engineering delivered modest improvements, while historical lag patterns remained the primary forecasting signal.

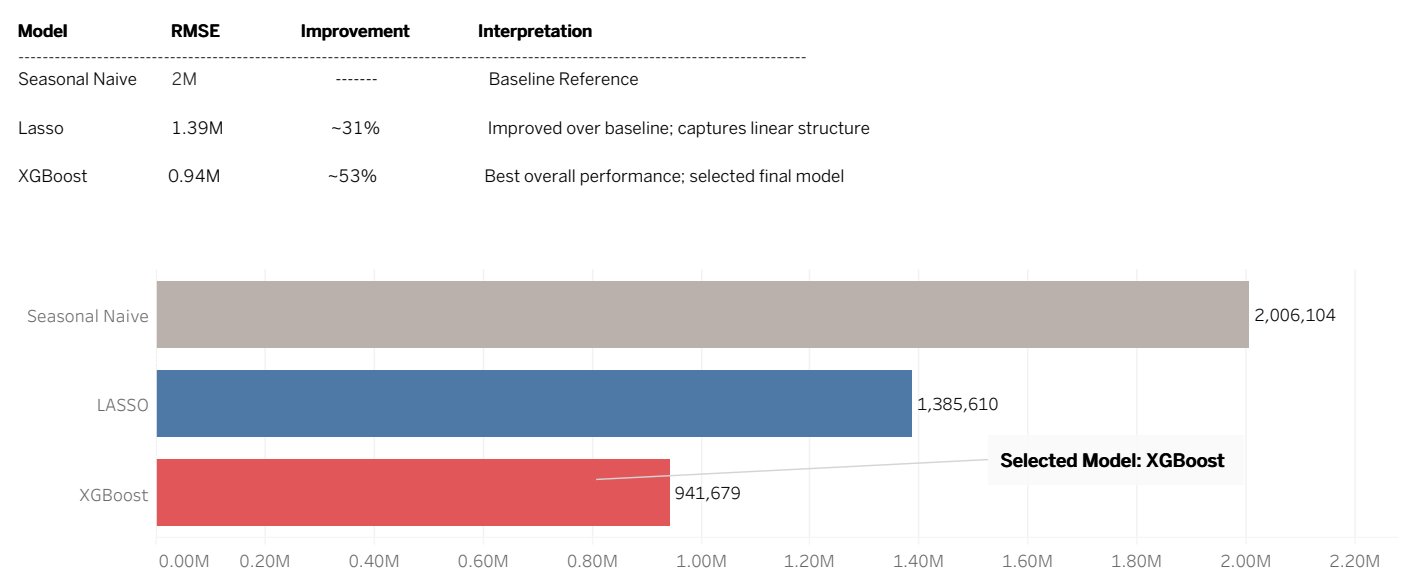
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Model Performance Comparison (RMSE — Lower Is Better)

Model Comparison & Selection	Performance
- Seasonal Naive used as benchmark - LASSO captured linear trend improvements - XGBoost achieved lowest RMSE - Selected as final forecasting model	Backtesting RMSE - Seasonal Naive: 2.01M - LASSO: 1.39M - XGBoost: 0.94M - Improvement vs Baseline: 53%



XGBoost best captured nonlinear demand variation and delivered the lowest forecast error.

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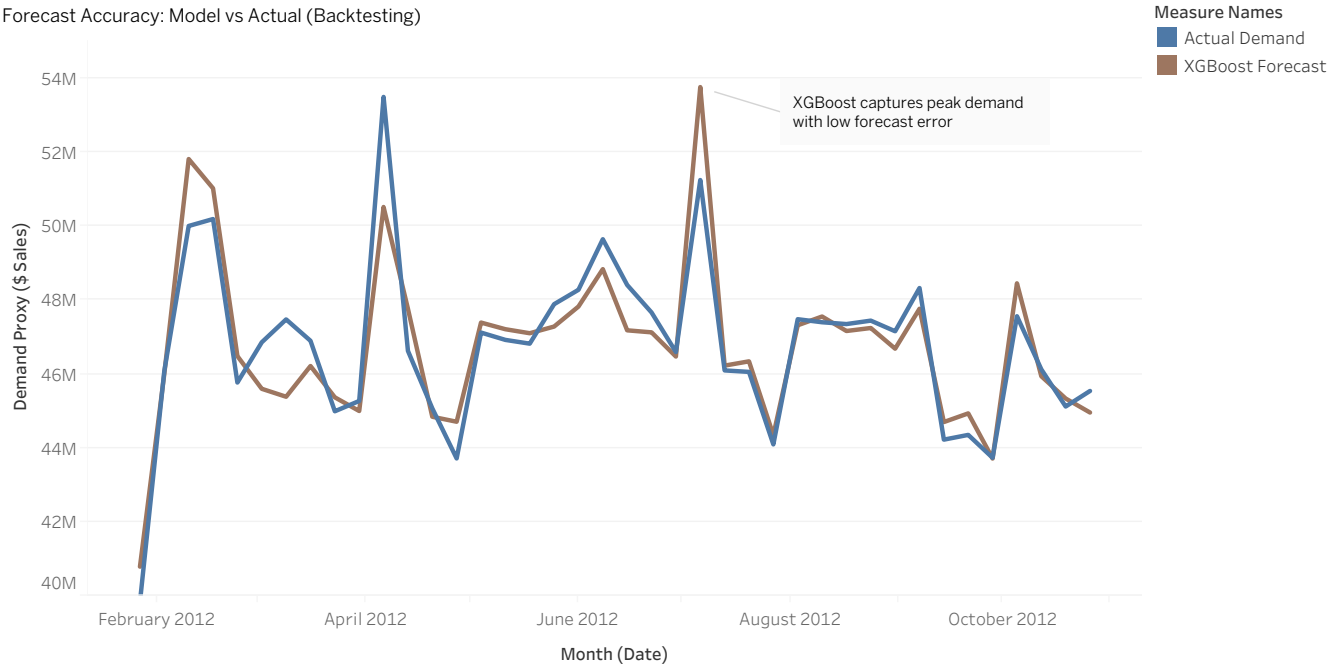
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Feature Engineering Enhances Forecast Performance	XGBoost delivers the most accurate forecasts	XGBoost delivers reliable and accurate forecasts over time	XGBoost Reduces Estimated Staffing Costs and Improves Planning Efficiency	Forecasts Support VET and VTO Staffing Decisions
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XGBoost Forecasts Remained Accurate on Unseen Data

- Model Validation Summary
- Forecast closely tracks actual weekly demand
 - Captures major seasonal spikes and dips
 - No persistent over/under-forecast bias
 - Stable performance on unseen holdout period
- Business Impact:
- Reliable VET / VTO trigger timing
 - Better staffing decisions with less guesswork

Forecast Accuracy: Model vs Actual (Backtesting)



XGBoost maintained high accuracy on unseen weeks, confirming readiness for operational forecasting.

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XGBoost delivers the most accurate forecasts

XGBoost delivers reliable and accurate forecasts over time

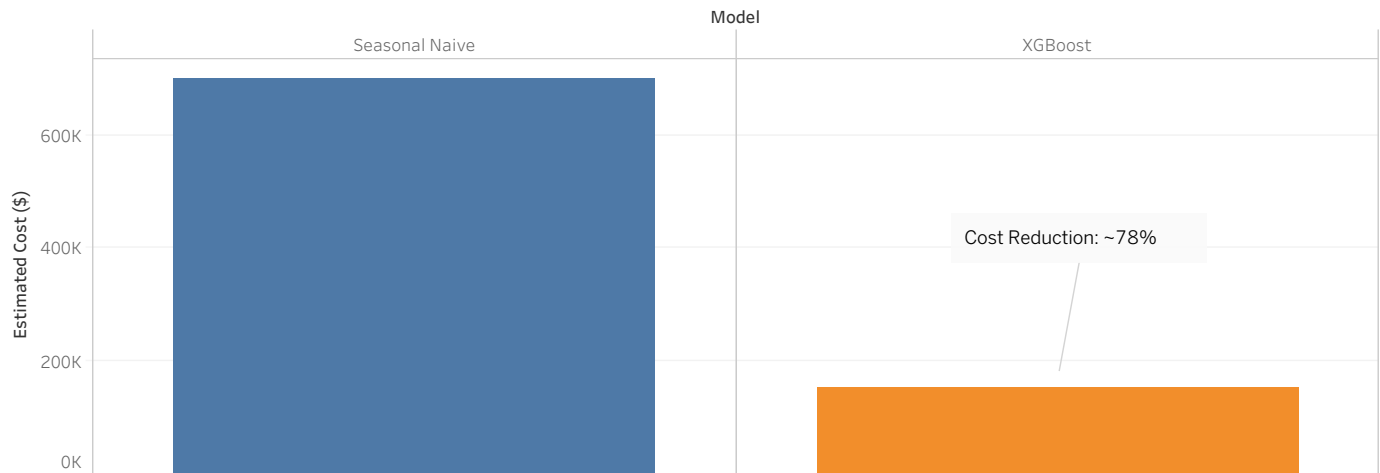
XGBoost Reduces Estimated Staffing Costs and Improves Planning Efficiency

Forecasts Support VET and VTO Staffing Decisions

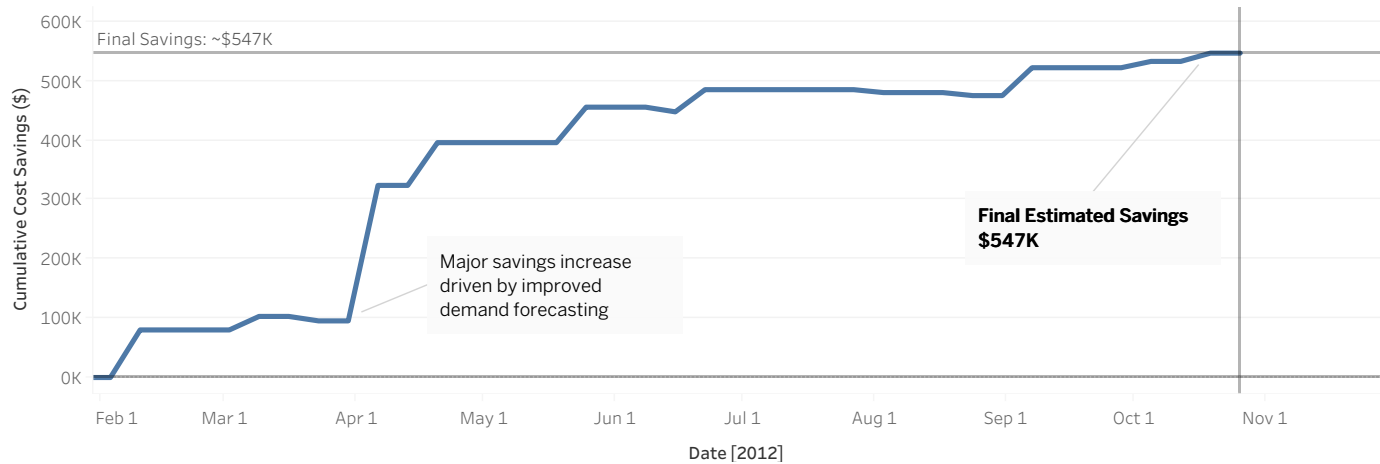
AI Decision Engine Converts Forecasts into Staffing Actions decisions

XGBoost Forecasting Delivered Significant Staffing Savings

Estimated Staffing Cost: Baseline vs XGBoost



Cumulative Savings Generated Over Holdout Period



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XGBoost delivers reliable and accurate forecasts over time

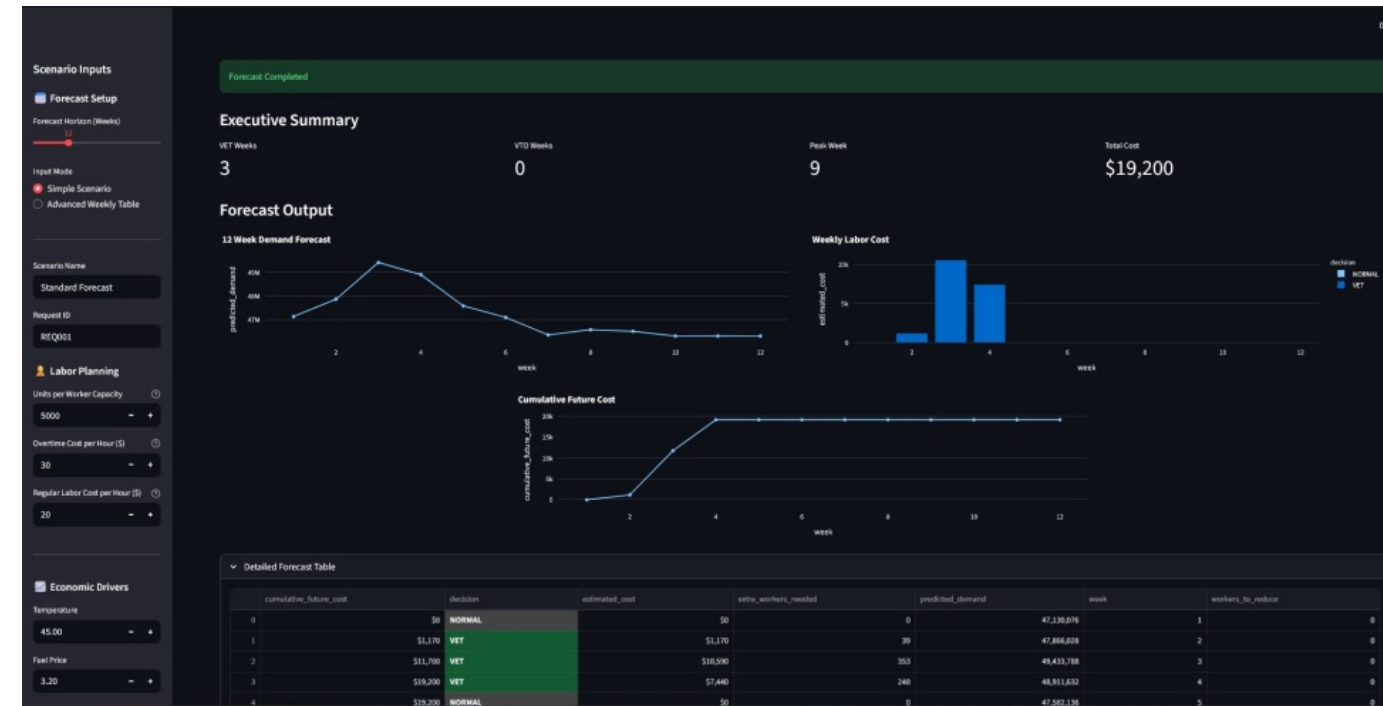
XGBoost Reduces Estimated Staffing Costs and Improves Planning Efficiency

Forecasts Support VET and VTO Staffing Decisions

AI Decision Engine Converts Forecasts into Staffing Actions decisions

Forecasting Improved Staffing Efficiency and Reduced Labor Cost

Forecasting Engine Automates VET & VTO Staffing Decisions
The system converts demand forecasts into labor recommendations and projected cost impacts.



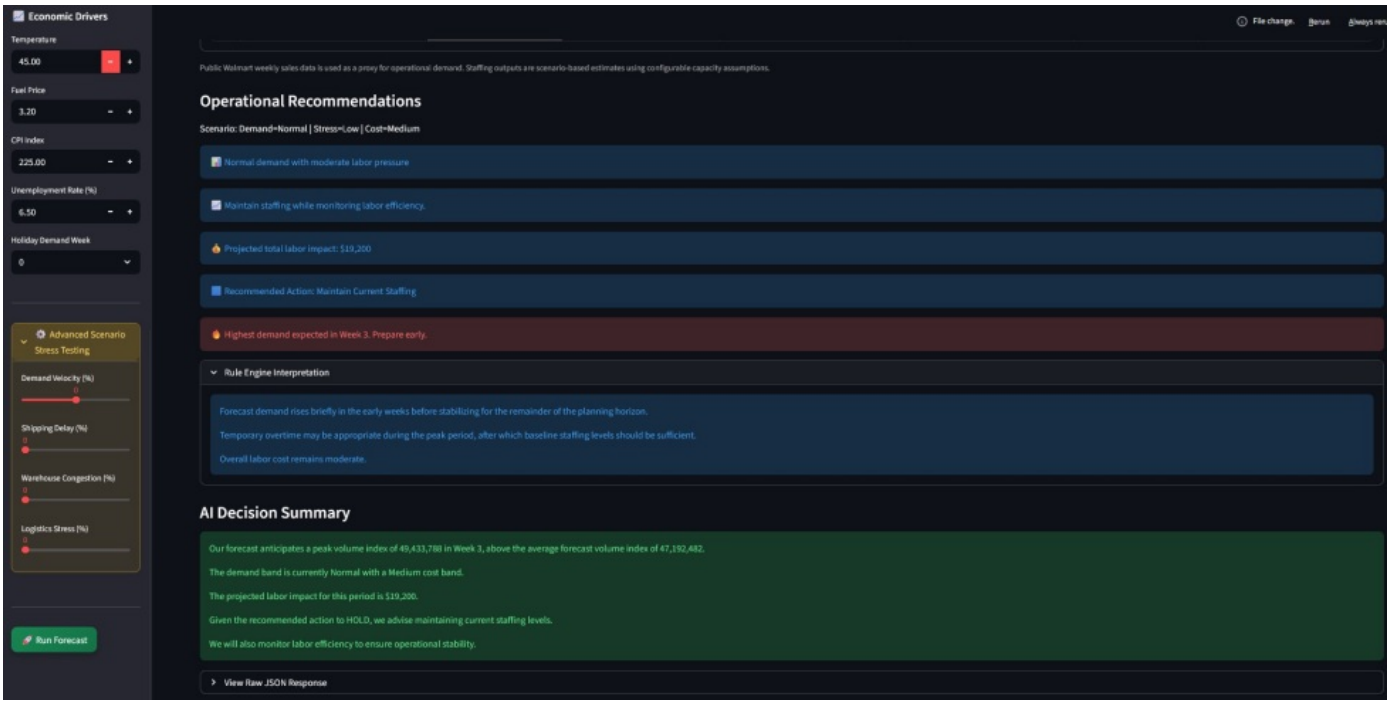
- Generates 12-week demand forecasts
- Converts demand into VET / VTO actions
- Supports proactive labor planning

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XGBoost Reduces Estimated Staffing Costs and Improves Planning Efficiency	Forecasts Support VET and VTO Staffing Decisions	AI Decision Engine Converts Forecasts into Staffing Actions decisions	Forecasting Improved Staffing Efficiency and Reduced Labor Cost
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Recommendation Engine Interprets Forecasts for VET / VTO Planning
Forecast outputs are translated into clear labor recommendations, cost estimates, and proactive staffing guidance...



- Interprets demand forecasts into VET / Normal / VTO actions
- Explains recommended staffing decisions
- Estimates projected labor impact
- Supports scenario stress testing
- Reduces manual planning effort

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Staffing efficiency improved and labor costs reduced

1. Key Result

- XGBoost delivered strongest forecast accuracy
- Estimated labor cost reductions identified
- Enabled proactive staffing planning

2. Operational Change

- Shifted from reactive staffing
- Moved toward forecast-driven workforce planning

3. Business impact

- Reduced overtime (VET)
- Reduced idle labor (VTO)
- Improved operational efficiency

4. Production Readiness & Scale

- Live cloud-hosted application already operational
- Scalable to multiple facilities
- Supports real-time planning decisions

Closing Statement

This project demonstrates how forecasting can convert demand uncertainty into proactive workforce decisions.

Live Application: (wake up: <https://warehouse-backend-n7on.onrender.com/>)
<https://warehouse-frontend-ce88.onrender.com/>

GitHub Repository:

<https://github.com/dracule99/VET-VTO-Forecasting>

LinkedIn Portfolio:

<https://www.linkedin.com/in/gammaconsult/>